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Investigating pedestrian suggestions for external features on fully autonomous vehicles: A virtual reality experiment

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ABSTRACT

The aim of this research was to identify potential external features on a fully autonomous vehicle (FAV) and investigate which features would help pedestrians to understand the intended behavior of a FAV at a crosswalk, improve their receptivity toward FAVs, and affect their crossing behavior. In the case of a FAV, technology may be primarily responsible for control of a vehicle, and therefore, interpersonal communication is not possible. The researchers wanted to identify potential interface/s on FAVs to make pedestrian-FAV interaction positive, in which pedestrians receive a clear message about the vehicle's intended action. In an experimental study, thirty participants walked across a virtual crosswalk in front of a FAV. Four visual and four audible features were tested. At the beginning of the study, the participants responded to a baseline receptivity survey, and at the end, they replied to a demographic questionnaire and a pedestrian behavior questionnaire, gave ratings for the features, and completed a personal innovativeness scale and an after-study receptivity survey. Crossing time and waiting time were collected from the simulator data. The results showed that pedestrians' receptivity toward FAVs significantly increased with the inclusion of external features. A walking silhouette or 'braking' in text were the most favored visual interfaces, while a verbal message was found to be the preferred audible feature. Females and people from 30+ age group reacted the most positively to the features. Those pedestrians who often commit errors or who show aggressive behaviors toward other road users rated the implementation of FAVs poorly, even with the external features. On the other hand, pedestrians who intentionally violate traffic rules and those who get distracted on the road were found to be more cautious in the presence of FAVs and appreciated the inclusion of the external features.

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1. Introduction

Once road-users enter the road network, they start a constant exchange of information with the traffic environment and other road-users around them in order to be ready to respond immediately. Drivers use various formal and, in some cases, mandatory methods to communicate with other road users, for example, turn signals, brake lights, and emergency lights. Along with these formal methods, many informal methods are often used to communicate with other road users: eye contact, facial expression, and hand gestures (Charisi et al., 2017; Šucha, 2014).

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Table 1
SAE proposed levels for vehicle automation. Source: SAE International (2014).

SAE levels for vehicle automation		Description
SAE Level 0	No automation	Human driver performs driving task even when equipped with warning and/or intervention systems
SAE Level 1	Driver assistance	Human driver performs driving task with the help of one driver assistance system of either steering or acceleration/deceleration
SAE Level 2	Partial automation	Human driver performs driving task with the help of one or more driver assistance systems of both steering and acceleration/deceleration
SAE Level 3	Conditional automation	Automated driving system performs mode-specific driving; however, the human driver must be ready to take back control to a request to intervene
SAE Level 4	High automation	Automated driving system performs driving tasks, even if a human driver does not need to take back control to a request to intervene. However, the automated system can operate only in certain environments and under certain conditions
SAE Level 5	Full automation	Automated system performs all driving tasks, in any environment and under all conditions that can be conducted by a human driver

In this era of advanced automated technology, automated vehicles at different levels (See Table 1) are expected to dominate the future transportation system. A fully autonomous vehicle (FAV) is categorized as level 5 vehicle automation technology that operates independently by means of software and hardware instead of a human driver and executes all safety-critical driving tasks for an entire trip. With a FAV, other than providing navigation input, human operators and/or passengers are not expected to take control of the vehicle at any time during the trip. While on the road, these vehicles will communicate with road users through formal means only; interpersonal interaction will not be possible. Among road users, this change will be hardest for pedestrians who are among the most vulnerable and who often assure their safety by interacting with vehicle operators while crossing a road using informal modes of communication; for example, eye contact or hand gestures (Llorca et al., 2011; Mirnig et al., 2017). Interpersonal interaction can offer a sense of road safety that technology is not currently able to provide. At least in the initial stages of FAV's implementation, pedestrians' sense of comfort may be challenged because of trust issues with this new and unproven technology. A survey study revealed that inability to establish interpersonal communication is one of the major reasons for an increase in perceived risk for the vulnerable road-users (pedestrians and bicyclists) when interacting with autonomous vehicles (Bikeleauge, 2014). Therefore, it is necessary to find a way for pedestrians to interact with a FAV. Recent research in this area has introduced different modalities of communication: visual intent displays or projections (Deb, Warner, Poudel & Bhandarib, 2016; Clamann, Aubert, & Cummings, 2017; Fridman et al., 2017; Mahadevan et al., 2018), auditory signals (Deb et al., 2016; Mahadevan et al., 2018), and physical prototypes using a robotic hand (Mahadevan et al., 2018) or humanoid robot (Mirnig et al., 2017). Most of these studies used picture and/or video-based surveys (Deb et al., 2016; Fridman et al., 2017) or real-world experiments (Clamann et al, 2017; Mahadevan et al., 2018). However, participants' exposures in a traffic environment with the threat of being hit by a vehicle may change their perceptions to a great extent. A real-world controlled experiment or a picture or video-based survey cannot create that threat.

This study used virtual reality (VR) to perform an experiment in a simulated traffic environment in which participants interacted with a FAV, equipped with external features, at a virtual crosswalk. The study was designed to (i) identify potential interface designs for FAV-pedestrian interaction and (iii) investigate which designs would allow pedestrians to understand the intended behavior of these vehicles at a crosswalk, improve their receptivity toward FAVs, and influence their crossing behavior accordingly. Deb, Carruth, Sween, Strawderman, and Garrison (2017) validated the VR platform used in this study for pedestrian safety research. The validation confirmed that pedestrians responded similarly in the VR traffic environment, as they would do in a real traffic environment. In order to keep participants only focused on the interface designs, the researchers chose to use FAV in this study, which can be operated by without any human operator. There was no option for interpersonal interaction with the vehicle, as the vehicle did not have any human operator.

2. Literature review

2.1. Interfaces for human-computer communication

Communication between computers and humans occurs through an interface designed to transmit a message from a computer to a human (Kammersgaard, 1988; Lancaster & Warner, 1993). The human, based on his/her recognition and understanding of the content of the message, validates its authenticity and interprets it as information or a request to perform a simple action. In the past, computer-human interfaces were not required in traffic for vehicle-pedestrian communication due to the presence of a human driver. With the advent of autonomous vehicles, this necessary communication must occur through designed interfaces. For consistency in the traffic system, interface designs supporting pedestrian safety and comfort should incorporate existing and familiar traffic components (signs, signals) or communication methods (smiles,

hand gestures) in a well-organized manner. In the field of human-computer interaction, new developments need to be easy to interpret, easy to use, and easy to learn, especially where hazards are present in the environment (Chen, Haas, & Barnes, 2007; Spool, 1999). Developing a potential human-computer interaction interface that can be used with minimal training (selecting existing interfaces used in pedestrian crossings rather than creating new ones), keeps task time short (in this case, time to make decision about crossing), and does not create high cognitive demands presents challenges (Issa & Isaias, 2015).

2.2. Current research on pedestrian-autonomous vehicle interaction

Traffic signs are not standardized internationally and signs in use today may be amended or modified in the future or new signs added. Also, pedestrians may move to a new place and share roads without further training or testing and still be expected to understand the signs and act on them properly (Ben-Bassat & Shinar, 2006; Ng & Chan, 2007). Similarly, when pedestrians encounter a FAV near a crosswalk for the first time, with no past experience with it or its external interfaces, they will need to be able to understand the messages being transmitted and act on them appropriately. It is important, therefore, that designers propose and evaluate many potential interface ideas for pedestrian-FAV interactions and consider options for improving their familiarity, compatibility, and standardization. Many research studies have recently proposed external features for FAVs to address the communication problem between pedestrian and FAV.

Lagstrom and Lundgren (2015) used a video-based approach and considered LED strips in different sequences to communicate the different modes of the vehicle (for example, 'about to start: LED strips shrinks down toward the center' or 'about to yield: LED strips expands toward the sides'). The results indicated that the pedestrians were able to understand the signals after only a short training and the interface replaced the informal communication of a human driver, with clearer and more prompt notifications. Nevertheless, these features do not provide a message about the vehicle's intention that is defined and understood by the general public (without previous training).

In 2016, using an online survey with 182 participants, Deb et al. identified pedestrians' expectations for external features on FAVs, considering both visual and audible features, and solicited participants' suggestions. Most of the respondents preferred a visual sign, such as a 'walking pedestrian sign' or a 'timer clock', indicating the vehicle's intention to stop at a crosswalk. The respondents also recommended including audible interacting features for distracted and visually impaired pedestrians. In another survey study, Fridman et al. (2017) tested 30 design interfaces for different states of an autonomous vehicle using responses from 200 participants. The study recommended using a green 'walk' in text with a pedestrian silhouette for safe crossing, while using 'do not walk' written in red and an upraised hand to stop pedestrians from crossing. The use of color alone may cause confusion based on different road-user perspectives. In a survey study collecting driver perception of external features on a FAV, Zhang et al. (2017) found that drivers of non-autonomous vehicles associated green with moving vehicles and red with a slowing or stopped vehicle; a forward flowing light with moving car and backward flowing light with a slowing or stopped car.

A Swedish firm Semcon tested their prototype displaying a smile as the external interface (Snyder, 2016). When this car detects a pedestrian, a smile lights up on the front car display confirming the car's intention to stop for the pedestrian. Prior research suggested that a smiling face on a car indicated its friendliness (Landwehr, McGill, & Herrmann, 2011). The researchers at Semcon also believed that pedestrians would feel comfortable crossing in front of a friendly smiling car. In another study, Clamann et al. (2017) tested a variety of designs for 'walk' and 'don't walk' signs. They concluded that pedestrians are more likely to base their road-crossing decisions on legacy behaviors (for example, the gap between them and the vehicle/s and the vehicle speed) rather than information presented on the external display. However, in this study, a human passenger was present in the driver seat to take control in case of an adverse situation. Studies using ghost drivers wearing seat costumes to make them invisible reported that pedestrians hesitated before crossing the road in front of the (apparently) autonomous vehicles (Rothenbücher et al., 2016; Sirkin et al., 2016). Therefore, the presence of a human driver in the Clamann et al. (2017) study may have confused the pedestrians regarding whether the controller of the vehicle was the human or the computer. This is also a significant concern with FAVs for other road users to understand who is in the control of the vehicle.

Using a different approach, the Imperial College of London and the Royal College of Art developed a new technology, called Blink (Peters, 2017). Blink detects a pedestrians' existence on the road and acknowledges them by showing a silhouette of a pedestrian using visual displays fitted to the four corners of the autonomous vehicle. If pedestrians want to cross and they're not at a crosswalk, they can hold up their hand, and the car will stop (provided it can brake in time) and then illuminate a green walk signal on the front display to let the pedestrians know that it is safe to cross. If they do not want to cross, they can wave the vehicle ahead, and it will signal by waving back that it understands and continue on its way. One criticism of this system would be that it puts the burden of interaction on the pedestrian: it requires they have some knowledge of the system and it depends on their ability to observe vehicles and make hand gestures.

The objectives of this study were to identify potential design features for FAV-pedestrian interaction, investigate pedestrian survey responses regarding their choice of the features and receptivity toward FAVs equipped with the features, and to analyze their crossing behavior in front of a FAV with features, when experienced for the first time. To better understand pedestrians' perception of external features, the researchers used a validated pedestrian simulator (Deb, Carruth, et al., 2017) which uses Unity 3D and a HTC Vive headset. The validation results from the Deb, Carruth, et al. (2017) study showed that the participants' walking speeds in the simulator matched the average pedestrian crossing rates in the published literature. In addition, the survey responses revealed that participants experienced a good sense of presence in the virtual

environment and rated the simulator with high points for usability and realism. The external features for the study were selected based on previous research outcomes, their feasibility for development, and pedestrians' ability to interpret them even without any prior knowledge or training on the feature designs. With a first time exposure to the vehicle and to the proposed features, researchers wanted to investigate whether a proposed feature carried a clear message to pedestrians about the vehicle's intention. In order to reduce the learning curve effect from a controlled experiment, only one trial condition (car approaching from the right and stops at the crosswalk) was used and a short list of features with diversified designs were considered. The features were green 'braking' in text from Deb et al. (2016), a smiling curve (Snyder, 2016), an image of a walking silhouette (Clamann et al., 2017; Deb et al., 2016), and audible features, such as music, a beep, and a verbal message saying 'safe to cross' (Deb et al., 2016; Urmson et al., 2015). In many situations, a vehicle may not need to come to a complete stop to let a pedestrian cross. Therefore, 'braking' was chosen over 'walk' in text indicating that the vehicle may slow down or stop based on the necessity of the situation. The researchers wanted to study the effect of external features for FAVs on pedestrians' choice for features and their receptivity toward FAVs at the first exposures to these vehicles. The researchers had to ensure that participants had looked for the vehicle and the feature for each of the trials before they started crossing and instructed the participants that the vehicle would stop only if it recognized them at the crosswalk. Therefore, proposed features indicated safe crossing conditions only and the car stopped at the crosswalk for each scenario.

3. Method

An experimental study was designed using a virtual reality environment. Potential external features for FAVs that would enhance pedestrian-FAV interactions were identified for this study. There were two modes of data collection: simulator-collected objective measures for pedestrian position and head rotation and survey-based responses on different subjective measures. The research was approved by the IRB of Mississippi State University and consent was obtained from each participant.

3.1. Participants

A total of 30 participants (17 males: mean age = 30.65, range = 22–47 and 13 females: mean age = 31.62, range = 18–47) were recruited from the communities surrounding Mississippi State University. This sample size is in line with similar studies investigating road-user behavior (Nagamatsu et al., 2011; Zito et al., 2015). Many driving simulator studies also used less than 30 participants in their study (Donmez, Boyle, & Lee, 2006; Itoh, Horikome, & Inagaki, 2013). None of the participants experienced simulation sickness and data from all 30 participants was used for analyses. The study was designed for U.S. citizens and all participants were fluent English speakers. The subjects had normal or corrected full-color vision and were able to walk at a normal pace and gait. The participants reported no hearing problems and only two had previous experience with virtual reality. The study took about 40 min to complete and participants received $20 in compensation for their time.

3.2. Pedestrian simulator

The pedestrian simulator used in this study was developed by Sween, Deb, Carruth, Waddell, and Furuichi (2016) and validated by Deb, Carruth, et al. (2017). The setup consisted of a virtual environment (VE), developed using Unity 5 3D software, and an HTC Vive VR headset with two Lighthouse sensors to track head movement. For this experiment, the Lighthouse sensors were positioned facing each other at opposite ends, approximately 8 m apart. The area tracked by the Vive was approximately 4 by 7 m with a 5.5-meter-long crosswalk and sidewalks on both sides.

3.3. Virtual environments

Three VEs were designed for the study: a familiarization VE, a lobby area, and a test environment. All VEs were implemented in Unity 5. The first VE was a general familiarization environment (shown in Fig. 1a), which allowed participants

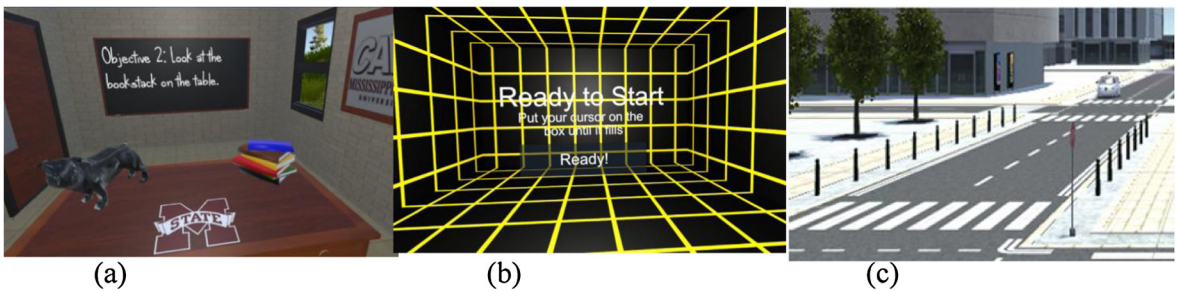

Fig. 1. Virtual environments: (a) Familiarization VE (b) Lobby area (c) Testing VE.

to become familiar with navigation in VR in general as well as to interact with the specific interface elements that would be used during the study. The second VE, the lobby area (shown in Fig. 1b), directed participants to the next trial and provided an opportunity to give instructions to them within the VE. A virtual footstep marker, placed on the ground in the lobby environment, indicated where the participants should stand. When the participants were standing on the marker, they were instructed to look at a virtual button to indicate that they were ready to continue.

The third VE was the actual test environment that replicated an urban downtown scene. The experimental trials took place at a pedestrian crosswalk and focused on a four-way intersection of two-lane streets with two-way stop signs (as shown in Fig. 1c). Participants started on a sidewalk facing one of the crosswalks with the intersection to their left.

3.4. Experimental design and measures

A randomized complete block design with 4 × 4 treatment combinations was used to investigate the effect of interface designs on pedestrian-FAV interaction. The participant was the block and the treatments were visual features (4 levels: a walking silhouette, 'braking' in text, a flashing smile, a control = no signal) and audible features (4 levels: a horn sound, music, a verbal warning saying 'safe to cross, a control = no signal). The visual features are displayed in Fig. 2. In order to control all other variables and focus on only pedestrians' choice of features, there was only one trial condition: a FAV approached from the farthest lane, perpendicular to the lane with stop signs, and stopped at the intersection. To avoid possible pedestrian-FAV collision, the vehicle was designed to approach only from the farthest lane and provided participants with enough time to make their crossing decision. As the proposed features indicated safe crossing conditions only, the FAV stopped in each of the trials at the crosswalk. No ambient traffic was included in this study to eliminate possible distractions and to ensure participants focus on and clear visibility of external features.

The researchers were interested to explore pedestrians' preferences for the feature or feature combinations based on their crossing behavior. For example, how long it took for the feature to convince them that it was safe to cross and how comfortable they felt or how much they trusted an autonomous vehicle to cross in front of it with a normal pace and gait. Therefore, dependent variables included objective measures for waiting time before crossing and crossing time, collected from the headset tracking data. A questionnaire (attached in the Appendix A) was used to collect subjective measures in terms of ratings for different interfaces. Each rating measured the pedestrians' feeling of safety for a proposed feature or feature combination on the FAV while crossing roads. To understand demographic effects, participants' age and gender were recorded. To explore the effect of individuals' personal opinions toward this new technological proposition (audible and visual interface on FAVs), personal innovativeness scores were calculated based on a 4-item personal innovativeness scale (adapted from Agarwal and Prasad (1998). Personal innovativeness is the general measure of the willingness to try something new in the face of unknowns (Agarwal & Prasad, 1998). This can influence an individual's attitude, behavior, and perception regarding a specific new idea (Chen & Chen, 2011; Lee, Qu, & Kim, 2007). Also, to investigate any change in general pedestrian behavior and receptivity of FAVs due to the installation of interfaces, pedestrian behavior scores and receptivity of FAVs scores were evaluated using a short version (20-item) of the pedestrian behavior questionnaire (PBQ, Deb, Strawderman, Carruth, et al., 2017) and two scenario-based questionnaires from the pedestrian receptivity questionnaire for FAVs (PRQF,

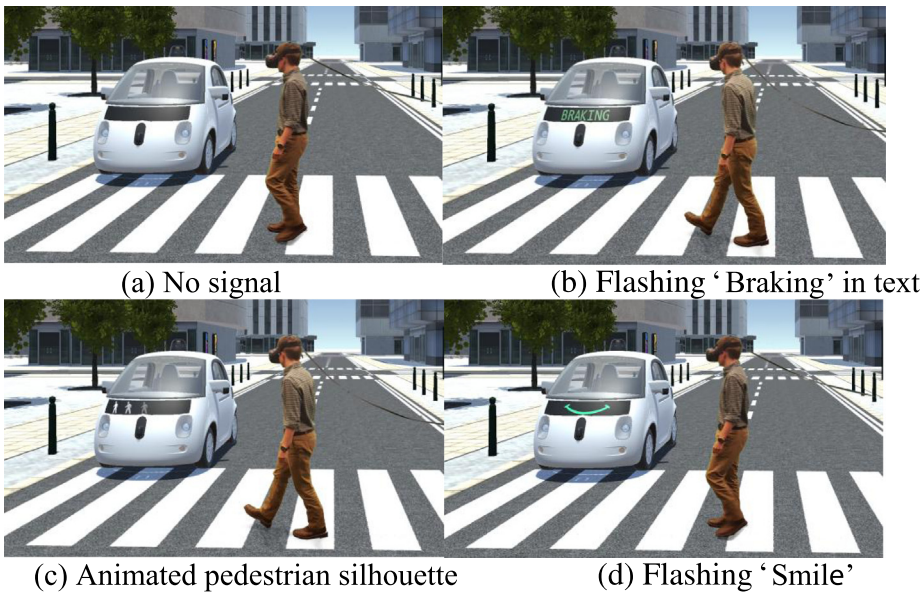

Fig. 2. Visual design interfaces for FAVs.

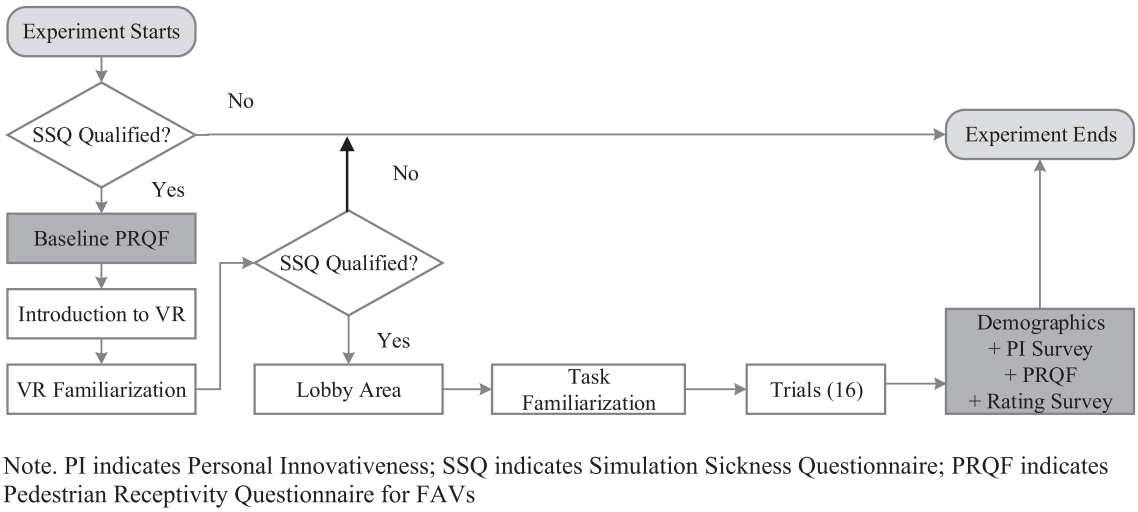

Fig. 3. An overview of the study procedure. Note. PI indicates Personal Innovativeness; SSQ indicates Simulation Sickness Questionnaire; PRQF indicates Pedestrian Receptivity Questionnaire for FAVs.

Deb, Strawderman, DuBien, et al., 2017). The PBQ measured five behavior scores for violation, error, lapse, aggressive behaviors, and positive behaviors. The two questionnaire scenarios from PRQF were designed to determine pedestrians' ability to trust FAVs enough to cross in front of them and their perception of the compatibility of FAVs with the existing traffic environment. In order to compare pedestrians' receptivity of FAVs with and without external features, the PRQF was administered before and after participants had encountered a FAV crossing the road in this experiment.

3.5. Procedure

The procedure of this study is drawn as a flowchart in Fig. 3. Upon arrival, the participants were asked to read and sign the consent form as a confirmation of their agreement to participate. Following the consent, they filled out a 16-item simulation sickness questionnaire (SSQ, Kennedy, Lane, Berbaum, & Lilienthal, 1993) to report their baseline health status. A score above 5 on the SSQ disqualified them from participation. No participants experienced simulation sickness in this study and therefore, were not disqualified. At this point, the qualified participants also responded to the 2-item scenario-based survey on their receptivity toward FAVs. The participants were then asked to put on the headset and explore the general-familiarization VE. After this step, they took off their headset and responded to the SSQ for the second time. Qualified participants (those with scores 5 and below) were selected to perform practice walks in the test environment. The practice walks included three crossings on the crosswalk that was specified for the study without any traffic in the environment.

In the experimental study, each of the participants went through sixteen randomly ordered trials (4×4). With the knowledge about FAV's safer operation, participants may assume that the car will always stop for them and they may immediately step in to the road without waiting for the vehicle and the features. As the purpose of the study was to evaluate pedestrians' preference for the features in terms of their travel behavior as well, it was necessary to make sure that they had acknowledged the features before they initiated crossing. Therefore, the participants were told that if the car perceives the participant at the crosswalk, it will come to a stop and they should cross when they felt it was safe to do so. By providing this information, the researchers motivated participants to look at the vehicle for its intended action (cues for deceleration). The FAV always came from the right and stopped at the intersection in each of the trials. Participants entered the lobby environment after each trial. In this environment, they were able to look around, ask questions, and pressed the 'Ready' button to continue with the next trial. After completion of the sixteen trials, the participants took off the headset and completed the demographic information survey (age, gender, hearing ability, visibility, and previous experience with VR), the PBQ, and the interface-rating survey, and again completed the 2-item scenario-based survey to determine after-experience pedestrian receptivity toward FAVs. With the completion of these surveys, the participants were compensated and the experiment was terminated.

4. Results

4.1. Influence of feature and feature combination on participants' ratings

The data was analyzed using SAS version 9.4 for the survey responses and the simulator-collected objective measures (crossing time and wait time before entering crosswalk). ANOVA (See Table 2) with feature ratings showed that the two

Table 2
Effect of features on feature ratings.

Factors	*df*	*F* Statistics	p value	Effect size (η^2)
Visual	(3, 463)	**34.02**	**<.0001**	0.151
Audible	(3, 463)	**19.48**	**<.0001**	0.086
Visual*audible	(9, 463)	**5.74**	**<.0001**	0.076
Participant (Block)	(1, 463)	2.91	0.0884	0.004

Note: Bold results are significant at $\alpha = 0.05$.

factors, audible and visual features, interacted significantly to affect the average rating for interface designs. Main effects for each of the treatments were also observed. Higher effect size for the main effect of visual features implies that participants' ratings for feature combination was influenced more by the inclusion of visual feature/s. In addition, the effect size of visual feature ($\eta^2 = 0.151$) is about twice as strong as the effect size for audible feature ($\eta^2 = 0.086$).

Further analysis using multiple comparisons ($\alpha = 0.05$) revealed that there was a significantly higher scoring for the ratings of the walking silhouette and text features as compared to the other two visual features. The verbal message was rated significantly higher compared to any other audible options. Overall, the FAV with no signal feature (neither visual nor audible) was the least-favored option of all, with a significantly lower rating of 1.97 out of 7. Effects for other feature combinations were not found to be significantly different. The multiple comparison results for two features and feature combinations are presented in Table 3 and 4 respectively.

4.2. Influence of feature and feature combination on crossing behavior

ANOVA of the objective measures revealed that the audible features significantly affected the crossing time. The effect size was calculated as to be $\eta^2 = 0.03$. No other significant effects were found for both objective measures: crossing time and waiting time. ANOVA results are presented in Table 5 and multiple comparison result for audible feature with respect to crossing time is presented in Table 6.

4.3. Influence of demographic variables

ANOVA was conducted to explore the influence of age [age 18–30: 13 people, 31–40: 9 people, and 40+: 8 people)] and gender [male: 17 and female: 13] on external features with respect to the feature ratings (see Table 7), and a regression analysis was conducted to identify the influence of personal innovativeness. The results showed an overall significant gender effect on pedestrian rating for features, although no significant interaction between gender and feature/s was found. After gaining experience with external interfaces on FAVs, female participants were found to be more receptive [M = 5.43, SD = 1.82] than males [M = 4.81, SD = 1.86] toward FAVs in terms of feature ratings.

On the other hand, age showed a significant main effect as well as a significant interaction effect with visual features on pedestrian rating for features. The main effect was found to be stronger than the interaction effect. Post hoc analysis with the three above-mentioned age groups revealed that the older age groups (31+) found it more necessary to have an external interface as compared to the younger age group (18–30). Nevertheless, young people (18–30) found the walking silhouette and the 'braking' signs significantly more trustworthy than the smile or no-visual feature. Participants aged 31+ also found the no-visual feature to be the worst condition for encouraging trust toward a FAV. They (age 31+ years) indicated that as long as there was a visual feature to notify them about the vehicle's intention, they would trust them. No interaction effect of age was found with audible features. There was no influence of personal innovativeness on the feature ratings. Table 8 shows the multiple comparison results for the interaction effect between age and visual features.

Effect of demographic factors on the objective measures are presented in Table 9. In the case of the objective measures from the simulator, gender was found to have significant influences on both crossing time and waiting time with a small effect sizes. Males surprisingly required longer waiting time and crossing time than females [waiting time: males ($M = 5.92$, SD = 0.086) and females ($M = 5.65$, SD = 0.035); crossing time: males ($M = 3.12$, SD = 0.095) and females ($M = 2.94$, SD = 0.15)]. Age was also found to be influential for both crossing time and waiting time. However, unlike with the survey responses, age did not interact with the audible or visual factors to affect the pedestrians' intended behavior in front of a FAV. Further analyses with multiple comparisons showed that all age groups exhibited significantly different mean crossing times [aged 18–30: $M = 5.31$, SD = 0.91; aged 31–40: $M = 5.69$, SD = 0.78; aged 40+: $M = 5.89$, SD = 1.00]. In the case of the waiting time, people aged 18–40 delayed significantly less than the people above 40 years old [aged 18–30: $M = 2.62$, SD = 0.77; aged 30–40: $M = 2.90$, SD = 1.60; aged 40+: $M = 5.89$, SD = 0.79]. Results from the regression analyses revealed that personal innovativeness has an influence on the waiting time [$\beta = -0.17$, $t(478) = 4.50$, $p < .0001$]: participants with a higher score on the personal innovativeness scale showed a lower waiting time before they started crossing the road in front of a FAV.

Table 3
Multiple comparison (Bonferroni) for main effects of audible and visual features for ratings.

Visual	Rating mean	Bonferroni grouping	Audible	Rating mean	Bonferroni grouping	
Silhouette	5.93	A A	Verbal	5.91	A	
Text	5.54	A	Beep	5.18	B B	
Smile	4.78	B	Music	4.79	B	C C
No Visual	4.05	C	No Audible	4.43		C

Note: Means with the same letter are not significantly different.

Table 4
Multiple comparison of interaction effects between audible and visual features for ratings.

Audible	Visual	Rating LSMEAN	LSMEAN grouping				
Verbal	Silhouette	6.47	A				
Verbal	Text	6.00	A	B			
Beep	Silhouette	5.93	A	B			
Verbal	None	5.83	A	B	C		
None	Silhouette	5.80	A	B	C		
Beep	Text	5.57		B	C		
Music	Silhouette	5.53		B	C		
None	Text	5.47		B	C	D	
Verbal	Smile	5.33	E	B	C	D	
Music	Text	5.13	E		C	D	F
Beep	Smile	4.70	E			D	F
Music	Smile	4.60	E		G		F
Beep	None	4.50			G		F
None	Smile	4.50			G		F
Music	None	3.90			G		
None	None	1.97					H

Note: Means with the same letter are not significantly different.

Table 5
ANOVA results for the effect of feature and feature combinations on the objective measures.

		Crossing time		Waiting time	
Factors	*df*	*F* Statistics	*p* value	*F* Statistics	*p* value
Visual	(3, 463)	0.80	0.4945	1.39	0.2457
Audible	(3, 463)	**2.87**	**0.0362**	0.08	0.9726
Visual*audible	(9, 463)	1.65	0.0978	0.74	0.6745
Participant (Block)	(1, 463)	2.56	0.1103	0.80	0.3723

Note: Bold results are significant at $\alpha = 0.05$.

Table 6
Multiple comparison (Bonferroni) for main effect of audible feature for crossing time.

Visual	Crossing time mean	Bonferroni grouping
Verbal	5.67	A
Music	5.77	A
No Audible	5.83	B
Beep	6.33	C

Note: Means with the same letter are not significantly different.

4.4. Influence of pedestrian general behaviors on FAV receptivity

Regression analysis was conducted to find the influence of pedestrians' general behaviors (based on PBQ subscales) on their ratings for the features and on the simulator-collected objective measures. Five different types of pedestrian behavior

Table 7
Effect of demographics on external feature ratings.

Effect	*F* stat. (*df*)	*p* value	η^2	Effect	*F* stat. (*df*)	*p* value	η^2
Gender	**19.26 (1, 448)**	**<0.0001**	**0.043**	Age	**16.21 (2, 432)**	**<0.0001**	**0.075**
Audible*gender	0.24 (3, 448)	0.8675	0.001	Audible*age	0.78 (6, 432)	0.5849	0.010
Visual*gender	2.05 (3, 448)	0.1064	0.014	Visual*age	**2.78 (6, 432)**	**0.0117**	**0.048**
Audible*visual*gender	0.09 (9, 448)	0.9997	0.002	Audible*visual*age	0.13 (18, 432)	0.9999	0.005

Note: Bold results are significant at $\alpha = 0.05$.

Table 8
Multiple comparison for the interaction effect between age and visual feature.

Age	Visual	Rate LSMEAN	LSMEAN grouping			
31–40	Silhouette	6.50			A	
					A	
40+	Text	6.30	B		A	
			B		A	
40+	Silhouette	6.15	B		A	C
			B		A	C
40+	Smile	5.85	B		A	C
			B	D		C
18–30	Silhouette	5.55	B	D		C
				D		C
18–30	Text	5.45		D		C
				D		C
31–40	Smile	5.36		D		C
				D		
31–40	Text	5.28		D		
31–40	None	4.53			E	
					E	
40+	None	4.25	F		E	
			F		E	
18–30	Smile	4.13	F		E	
			F			
18–30	None	3.72	F			

Note: Means with the same letter are not significantly different.

Table 9
Effect of demographics on pedestrian crossing behavior.

Crossing time				Waiting time			
Effect	*F* stat. (*df*)	*p* value	η^2	Effect	*F* stat. (*df*)	*p* value	η^2
Gender	**5.44 (1, 448)**	**0.0201**	**0.012**	Gender	**5.59 (1, 448)**	**0.0185**	**0.013**
Audible*gender	0.06 (3, 448)	0.9820	0.000	Audible*gender	0.30 (3, 448)	0.8252	0.002
Visual*gender	0.17 (3, 448)	0.9182	0.001	Visual*gender	1.48 (3, 448)	0.2199	0.009
Audible*visual*gender	0.38 (9, 448)	0.9464	0.007	Audible*visual*gender	0.36 (9, 448)	0.9522	0.007
Age	**6.71 (1, 432)**	**0.0014**	**0.032**	Age	**11.20 (2, 432)**	**<0.0001**	**0.052**
Audible*age	0.39 (6, 432)	0.8842	0.005	Audible*age	0.45 (6, 432)	0.8439	0.006
Visual*age	0.78 (6, 432)	0.5898	0.011	Visual*age	0.57 (6, 432)	0.7515	0.008
Audible*visual*age	0.21 (18, 432)	0.9998	0.009	Audible*visual*age	0.68 (18, 432)	0.8308	0.028

(violation, error, lapse, aggressive behavior, and positive behavior) were considered as independent variables. The results, shown in Table 10, revealed a significant positive association between feature rating and positive behaviors, but significant negative association with errors and aggressive behaviors.

The results from the regression analyses, presented in Table 11, showed a significant influence of general pedestrian behaviors on crossing behaviors in front of a FAV. Pedestrians' errors and aggression positively influenced their crossing time, while their violations and lapses had negative influences on crossing time. For waiting time, violation and positive behavior had a positive influence on waiting time, but error and aggression negatively influenced waiting time.

4.5. Improvement in pedestrian receptivity

T-test analyses were conducted on ratings before and after exposure to the FAV with features, gathered using two scenario-based survey responses. The results showed that inclusion of features improved participants' receptivity level

Table 10
Effect of pedestrian behaviors on external feature preference.

Pedestrian behaviors	Parameter estimate (β)	t Value	*p* Value
Violation	−0.08547	−1.46	0.1443
Error	−0.15294	**−2.54**	**0.0112**
Lapse	−0.04661	−0.86	0.3899
Aggression	−0.23754	**−5.26**	**<0.0001**
Positive	0.12390	**2.75**	**0.0062**

Note. Bold results are significant at $\alpha = 0.05$.
df = 29.

Table 11
Influence of pedestrian behaviors on simulator collected objective measures.

Variable	Parameter Estimate (β)	t Value	Pr > |t|
Crossing time			
Violation	−0.11088	**−2.56**	**0.0108**
Error	0.20408	**4.57**	**<0.0001**
Lapse	−0.16860	**−4.19**	**<0.0001**
Aggression	0.15336	**4.57**	**<0.0001**
Positive	0.01656	0.50	0.6205
Waiting time			
Violation	0.27671	**6.63**	**<0.0001**
Error	−0.31012	**−7.23**	**<0.0001**
Lapse	−0.02975	−0.77	0.4419
Aggression	−0.16070	**−4.98**	**<0.0001**
Positive	0.20851	**−6.49**	**<0.0001**

Note. Bold results indicate significant effect.
df = 958.

toward FAVs. Participants were significantly more comfortable ($t = 2.21$, $df = 29$, $p < 0.0349$) interacting with the FAVs and crossing roads in front of them when the FAVs had an external interface installed. Also, the participants felt significantly more positive ($t = 4.27$, $df = 29$, $p < 0.0002$) about accepting these vehicles in their area if the features were included.

5. Discussion

The implementation of VR for conducting research in autonomous vehicle interactions allowed the participants to gain experience interacting with a driverless vehicle equipped with potential interfaces before they exist on the market. Based on this study, it appears that participants will respond to traffic in a simulator as they would in the real world; they look for the vehicle before crossing and wait for the vehicle to stop. The average walking speed for the pedestrians in front of an autonomous vehicle equipped with external features was 1.54 m/sec; faster than the general walking speed 1.20 m/sec (FHWA, 2009). Being in front of a vehicle without a human driver could be the cause for participants becoming unsure and thus walking faster. Overall, it can be said that this VR setting works well and can be a valuable mode for similar research involving FAVs and road users, especially pedestrians (Deb, Carruth, et al., 2017).

The study was designed to investigate pedestrians' choice of external features to improve their receptivity toward them as well as to understand their crossing behavior in front of a FAV with external features. The results revealed many interesting outcome on pedestrians' rating for different features, their crossing behavior in front of a FAV, influence of pedestrians' demographics and general behavior on their rating and crossing behavior, and the overall improvement in receptivity with the inclusion of external features. The following sections provide detailed discussions on the significant outcomes of the study.

5.1. Effect on feature rating

Several important findings emerged from the current study. The results confirm the necessity of including external interfaces on FAVs in order to ensure convenient pedestrian-FAV interaction. In accordance with some of the previous research in this area (Clamann et al., 2017; Deb et al, 2016; Fridman et al., 2017), text and walking silhouette were the most favored visual features as compared to the smiling curve while verbal message was the most favored audible feature. There are a multitude of things that can distract a pedestrian at an intersection (including texting/phone conversation), and an audible message has a better chance of breaking into that distraction. In addition, a verbal message would be useful for pedestrians with visual disabilities or in poor visibility conditions (Deb et al., 2016). On the other hand, participants might feel worried

that they would become confused by the beep or the music with all the other sounds that are part of the traffic environment and thus rated both the 'animated pedestrian silhouette' and the 'flashing 'braking' in text' with higher ratings in the presence of a verbal message (≥6). The walking silhouette is a very common feature at pedestrian crossings, and thus provided the pedestrians with something familiar, something they were comfortable with, as a signal of the vehicle's intention to let them cross. The participants also found the flashing text message clear enough to understand the FAV's intention to stop for them. However, the smile proved to be a less effective feature; the participants did not get a clear message about the vehicle's intended action. They were confused as to whether the car was thanking them for not crossing the road or was trying to convince them to cross.

5.2. Effect on crossing behavior

Clamann et al. (2017) study found that pedestrians still focus on vehicle speed and distance when making crossing decision. However, researchers of this study found that verbal message provided the most confidence for crossing the road without hesitation in the shortest time. Contrary to the survey ratings, music produced results similar to verbal messages in the simulator data. On the other hand, beep confused participants to such an extent that they hesitated to cross the road even longer than when no feature was present. The beep, which sounds like an alarm, was probably perceived as a warning instead of the notification for safe crossing when coupled with these two visual features. Both audible and visual signals were presented to the participants at the same time of each trial. Pedestrians were able to recognize the visual signals when they looked at them and understood them, however, visual signals were acknowledged at the time they were played. This could be the reason for audible signals having significant influence on the crossing time based on the message carried by them.

5.3. Demographic effect

It should be noted that the females rated the features with significantly higher ratings than the males. In general, females show lower receptivity toward technology, and this was also found to be true for FAVs and related automated technologies (Canada & Brusca, 1991; Payre, Cestac, & Delhomme, 2014). However, in this case, the inclusion of the external interfaces may have given female participants a perception of safety for themselves and for their friends and families that raised their receptivity toward the entire technology.

In general, younger age groups are more receptive to FAVs than are older populations, as would be expected. Young people are usually more positive about the long-term impact of technological change on life (Smith, 2014; Veruggio, Operto, & Bekey, 2016). For the younger generations, the interacting features were likely not necessary for FAV receptivity. On the other hand, inclusion of external features were necessary for the older population (30+). However, the demographic analyses for this study showed that the older age groups (30+) were more receptive to FAVs that were equipped with interacting features than the younger generations (18–30) were. Nevertheless, the specific choices of the younger people in favor of the walking silhouette and the 'braking' signs still indicated their desire for a clear message on the FAV display. Participants aged 31 to 40 found visual external features to be necessary on FAVs and preferred the familiar image of a walking silhouette; they did not trust the smile or the 'braking' in text options. After the experiment, many of those in this age group expressed their concern about getting confused by a smile or text as 'Braking'. They found the animated silhouette to be a clear and excellent signal indicating a safe walking condition. People aged 40+ years were more comfortable when there was a visual feature to notify them about the FAV's intention; they also stated that they would feel more certain or secure about the message on the FAV if it were flashing or animated.

5.4. Pedestrian behavior effect

The influence of pedestrian behaviors on their rating for designs showed that pedestrians who often commit errors or those who get aggressive on the road rated the features with lower scores. Errors are usually the result of a lack of knowledge about traffic rules, and aggressiveness is an individual characteristic. These people took a longer time to cross the road in front of the stopped FAV and waited less before entering the crosswalk. The presence of FAVs, even with the external features, would not benefit those pedestrians who do not have enough knowledge about the traffic system or who always expect to have the right of the way.

On the other hand, pedestrians who violate traffic rules intentionally and those who get distracted on the road were found to be more cautious about making a crossing decision in the presence of FAVs. They delayed before starting to cross and then crossed hurriedly to avoid being in front of a FAV longer than necessary. Those who violate traffic rules intentionally might be cautious knowing that FAVs act according to the rules, thus requiring them to follow the rules as well. In the case of distracted pedestrians, they usually take longer to cross roads (Schabrun, van den Hoorn, Moorcroft, Greenland, & Hodges, 2014). However, the presence of FAVs made them uncomfortable; as an unknown, they could not factor the FAV into their distracted walking and so laid aside their distraction in order to cross. These positive changes in pedestrians' general behavior due to the inclusion of FAVs can be mentioned as an enhancement in traffic safety. Pedestrians who frequently follow positive behaviors were also found to comply with rules for FAVs; they took their time to observe and understand the interacting interfaces before they started crossing.

Finally, the inclusion of interfaces on the FAVS was tested against FAVs without any interfaces and the results showed significant improvement in pedestrians' receptivity toward FAVs, both in terms of their crossing decisions as well as with their perception of FAVs compatibility in the existing traffic environment. This finding confirms the efficacy of including external features on FAVs.

5.5. Limitations

The experimental study was designed for a non-signalized intersection and involved a single controlled scenario in which a FAV came from one side of the road and stopped at the intersection. In order to remove any distraction and make the participants concentrate only on the features, no other traffic was added in the virtual environment. The proposed features for this study indicated safe crossing conditions only and therefore, the vehicle stopped in each of the trials. The scenario did not consider the termination of the feature so that participants would know when the vehicle would restart. This was the first exposure for the participants to a FAV and FAV with features. Making a complex design with too many varying factors may cause distractions and take their focus away from or visibility restricted from the features. The addition of more complex traffic situations with signals, vehicle operator, positive and negative interfaces, and environmental sounds could cause changes in a pedestrian's choice of features as well as in their intended behaviors. Future studies should consider a complex traffic system with presence of an operator and pedestrian signals at the crosswalk along with near miss conditions to observe pedestrian perception of some of the highly rated features from this study. The start and termination of signals at an intersection would also be important to investigate in future studies.

6. Conclusion

In recent years, transportation researchers and vehicle manufacturers have become increasingly enthusiastic about commercializing FAVs and entering them into the consumer market. FAVs can potentially enhance traffic safety by reducing human errors. However, the introduction of FAVs is likely to cause a large change in our way of life. Public approval of FAVs will not be possible overnight. Researchers should continue to conduct acceptance and interface design studies for identifying road-users' suggestions to help ensure FAV's successful implementation in the existing traffic system. Within these studies, the perspective of all road users must be considered.

The current study allowed pedestrians to experience interaction with a FAV in a simulated environment. For all the study participants, this was their first encounter of any kind with a FAV. The external displays were included on FAV to investigate which design/s can increase pedestrians' receptivity in FAVs at crosswalks and thereby influence their crossing behavior. The results showed that a familiar sign for pedestrian signals (walking silhouette), a clear text ('Braking') and a clear verbal message were significantly favored by the participants as compared to other proposed features. Their crossing behavior was mostly influenced by the audible signals, verbal message and music. Overall, the inclusion features on a FAV, audible or visual, significantly increased pedestrians' receptivity. Manufacturers and transportation researchers should consider these results and design the FAVs accordingly. Future studies can investigate the feasibility and preference for the reported features from other road-users' perspectives.

Appendix A. Feature rating scale

Based on your experience, how safe would you feel entering the crosswalk with the addition of following features to a fully autonomous vehicle?

1.	Walking sign						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
2.	'Braking' written in text						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
3.	Smiley face						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
4.	Horn sound						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
5.	Music						
	1	2	3	4	5	6	7
	Not safe at all						Very safe

6.	Verbal message						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
7.	Walking sign and horn sound						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
8.	Walking sign and music						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
9.	Walking sign and verbal message						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
10.	Braking in text and horn sound						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
11.	Braking in text and music						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
12.	Braking in text and verbal message						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
13.	Smiley face and horn sound						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
14.	Smiley face and music						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
15.	Smiley face and verbal message						
	1	2	3	4	5	6	7
	Not safe at all						Very safe
16.	No feature						
	1	2	3	4	5	6	7
	Not safe at all						Very safe

Note: Higher score for a feature or feature combination indicates greater preference for that feature option.

Short version of the PBQ (20 items)

Violations

V1 I cross the street even though the pedestrian light is red.
V2 I cross diagonally to save time.
V3 I cross outside the pedestrian crossing even if there is one (crosswalk) less than 50 m away.
V10 I take passageways forbidden to pedestrians to save time.

Errors

E2 I cross between vehicles stopped on the roadway in traffic jams.
E7 I cross even if vehicles are coming because I think they will stop for me.
E8 I walk on cycling paths when I could walk on the sidewalk.
E9 I run across the street without looking because I am in a hurry.

Lapses

L5 I realize that I have crossed several streets and intersections without paying attention to traffic.
L6 I forget to look before crossing because I am thinking about something else.
L7 I cross without looking because I am talking with someone.
L8 I forget to look before crossing because I want to join someone on the sidewalk on the other side.

Aggressive behaviors

A1 I get angry with another road user (pedestrian, driver, cyclist, etc.), and I yell at him.
A3 I cross very slowly to annoy a driver.

(*continued on next page*)

A4	I get angry with another road user (pedestrian, driver, cyclist, etc.), and I make a hand gesture.
A6	I have gotten angry with a driver and hit their vehicle.
Positive behaviors	
P1	I thank a driver who stops to let me cross.
P3	When I am accompanied by other pedestrians, I walk in single file on narrow sidewalks so as not to bother the pedestrians I meet.
P4	I walk on the right-hand side of the sidewalk so as not to bother the pedestrians I meet.
P5	I let a car go by, even if I have the right-of-way, if there is no other vehicle behind it.

V indicates Violations.
E indicates Errors.
L indicates Lapses.
A indicates Aggressive Behaviors.
P indicates Positive Behaviors.

Scenario-based question

You are walking home from shopping. On your way, you need to cross multiple crosswalks, both signalized and un-signalized. As you prepare to cross at an un-signalized crosswalk, you find that a driverless vehicle is approaching the crosswalk (with no one sitting in the driver seat). Based on the scenario, please select the choice that best reflects your behavior for each question.

A.	**What will your response at the crosswalk be, with the FAV approaching?**
1.	I will not cross the road at the crosswalk to avoid crossing in front of the FAV.
2.	I will run across the road even though the driverless vehicle has stopped for me.
3.	I will make sure that the driverless vehicle stops before I start crossing.
4.	I will wait to see if the vehicle decelerates before I start crossing.
5.	I will cross the road with full confidence that the driverless vehicle will stop for me.
B.	**As a pedestrian, how will you accept the presence of driverless vehicles in your area?**
1.	I will be angry to see driverless vehicles in my area; I think they will cause more problems.
2.	I will feel anxious about the presence of driverless vehicles in my area; I do not trust them.
3.	I will be indifferent to the presence of driverless vehicles in my area; it does not matter to me.
4.	I will have no problem with driverless vehicles in my area; I trust the technology.
5.	I will feel excited to see driverless vehicles in my area; I believe they will make my area safer.

References

Agarwal, R., & Prasad, J. (1998). A conceptual and operational definition of personal innovativeness in the domain of information technology. *Information Systems Research, 9*(2), 204–215.

Ben-Bassat, T., & Shinar, D. (2006). Ergonomic guidelines for traffic sign design increase sign comprehension. *Human Factors, 48*(1), 182–195.

Bikeleauge. (2014). Autonomous and connected vehicles: Implications for bicyclists and pedestrians. Retrieved from http://bikeleague.org/sites/default/files/Bike Ped Connected Vehicles.pdf. [Date accessed 2017-06-3].

Canada, K., & Brusca, F. (1991). The technological gender gap: Evidence and recommendations for educators and computer-based instruction designers. *Educational Technology Research and Development, 39*(2), 43–51.

Charisi, V., Habibovic, A., Andersson, J., Li, J., & Evers, V. (2017). Children's views on identification and intention communication of self-driving vehicles. In *Proceedings of the 2017 Conference on Interaction Design and Children* (pp. 399–404). ACM.

Chen, C. F., & Chen, P. C. (2011). Applying the TAM to travelers' usage intentions of GPS devices. *Expert Systems with Applications, 38*(5), 6217–6221.

Chen, J. Y., Haas, E. C., & Barnes, M. J. (2007). Human performance issues and user interface design for teleoperated robots. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews), 37*(6), 1231–1245.

Clamann, M., Aubert, M., & Cummings, M. L. (2017). Evaluation of Vehicle-to-Pedestrian Communication Displays for Autonomous Vehicles (No. 17-02119).

Deb, S., Carruth, D. W., Sween, R., Strawderman, L., & Garrison, T. M. (2017). Efficacy of virtual reality in pedestrian safety research. *Applied Ergonomics, 65*, 449–460.

Deb, S., Strawderman, L., Carruth, D. W., DuBien, J., Smith, B., & Garrison, T. M. (2017). Development and validation of a questionnaire to assess pedestrian receptivity toward fully autonomous vehicles. *Transportation Research Part C Emerging Technologies, 84*, 178–195.

Deb, S., Strawderman, L., DuBien, J., Smith, B., Carruth, D. W., & Garrison, T. M. (2017). Evaluating pedestrian behavior at crosswalks: Validation of a pedestrian behavior questionnaire for the US population. *Accident Analysis & Prevention, 106*, 191–201.

Deb, S., Warner, B., Poudel, S., Bhandari, S. (2016). Identification of external design preferences in autonomous vehicles. In: Proceedings of the 2016 industrial and systems engineering research conference. Anaheim, California.

Donmez, B., Boyle, L. N., & Lee, J. D. (2006). The impact of distraction mitigation strategies on driving performance. *Human Factors, 48*(4), 785–804.

Federal Highway Administration (FHWA) (2009). Manual on Uniform Traffic Control Devices (MUTCD). Retrieved from http://mutcd.fhwa.dot.gov/kno_2009r1r2.htm.

Fridman, L., Mehler, B., Xia, L., Yang, Y., Facusse, L. Y., & Reimer, B. (2017). To Walk or Not to Walk: Crowdsourced Assessment of External Vehicle-to-Pedestrian Displays. arXiv preprint arXiv:1707.02698.

Issa, T., & Isaias, P. (2015). Usability and human computer interaction (HCI). In *Sustainable Design* (pp. 19–36). London: Springer.

Itoh, M., Horikome, T., & Inagaki, T. (2013). Effectiveness and driver acceptance of a semi-autonomous forward obstacle collision avoidance system. *Applied Ergonomics, 44*(5), 756–763.

Kammersgaard, J. (1988). Four different perspectives on human–computer interaction. *International Journal of Man-Machine Studies, 28*(4), 343–362.
Kennedy, R. S., Lane, N. E., Berbaum, K. S., & Lilienthal, M. G. (1993). Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *International Journal of Aviation Psychology, 3*(3), 203–220.
Lagstrom, T., & Lundgren, V. M. (2015). AVIP-Autonomous vehicles interaction with pedestrians [Doctoral dissertation, Thesis].
Lancaster, F. W., & Warner, A. J. (1993). *Information Retrieval Today. Revised, Retitled*. 1110 North Glebe Rd., Suite 550, Arlington, VA 22201: Information Resources Press.
Landwehr, J. R., McGill, A. L., & Herrmann, A. (2011). It's got the look: The effect of friendly and aggressive "facial" expressions on product liking and sales. *Journal of Marketing, 75*(3), 132–146.
Lee, H. Y., Qu, H., & Kim, Y. S. (2007). A study of the impact of personal innovativeness on online travel shopping behavior—A case study of Korean travelers. *Tourism Management, 28*(3), 886–897.
Llorca, D. F., Milanés, V., Alonso, I. P., Gavilán, M., Daza, I. G., Pérez, J., & Sotelo, M. Á. (2011). Autonomous pedestrian collision avoidance using a fuzzy steering controller. *Intelligent Transportation Systems, IEEE Transactions on, 12*(2), 390–401.
Mahadevan, K., Somanath, S., & Sharlin, E. (2018). Communicating awareness and intent in autonomous vehicle-pedestrian interaction. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 429). ACM.
Mirnig, N., Perterer, N., Stollnberger, G., & Tscheligi, M. (2017). Three strategies for autonomous car-to-pedestrian communication: A survival guide. In *Proceedings of the Companion of the 2017 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 209). 210ACM.
Nagamatsu, L. S., Voss, M., Neider, M. B., Gaspar, J. G., Handy, T. C., Kramer, A. F., & Liu-Ambrose, T. Y. (2011). Increased cognitive load leads to impaired mobility decisions in seniors at risk for falls. *Psychology and Aging, 26*(2), 253.
Ng, A. W., & Chan, A. H. (2007). The guessability of traffic signs: Effects of prospective-user factors and sign design features. *Accident Analysis & Prevention, 39*(6), 1245–1257.
Payre, W., Cestac, J., & Delhomme, P. (2014). Intention to use a fully automated car: Attitudes and a priori acceptability. *Transportation research part F: Traffic Psychology and Behaviour, 27*, 252–263.
Peters, A. (2017). What if driverless cars let you cross the street when you wave at them? Retrieved from https://wwwfastcompany.com/3068166/what-if-driverless-cars-let-you-cross-the-street-when-you-wave-at-them.
Rothenbücher, D., Li, J., Sirkin, D., Mok, B., & Ju, W. (2016). Ghost driver: A field study investigating the interaction between pedestrians and driverless vehicles. In *Robot and Human Interactive Communication (RO-MAN), 2016 25th IEEE International Symposium on* (pp. 795–802). IEEE.
SAE International. (2014). Automated driving: levels of driving automation are defined in new SAE international standard J3016.
Schabrun, S. M., van den Hoorn, W., Moorcroft, A., Greenland, C., & Hodges, P. W. (2014). Texting and walking: Strategies for postural control and implications for safety. *PLoS One, 9*(1), e84312.
Sirkin, D., Baltodano, S., Mok, B., Rothenbücher, D., Gowda, N., Li, J., ... Ju, W. (2016). Embodied design improvisation for autonomous vehicles. In *Design thinking research* (pp. 125–143). Springer International Publishing.
Smith, A. (2014). US Views of Technology and the Future. Pew Internet.
Snyder, J. B. (2016). This self-driving car smiles at pedestrians. Retrieved from https://www.autoblog.com/2016/09/16/this-self-driving-car-smiles-at-pedestrians/.
Spool, J. M. (1999). *Web site usability: A designer's guide*. Morgan Kaufmann.
Šucha, M. (2014). *Road users' strategies and communication: Driver-pedestrian interaction*. Transport Research Arena (TRA).
Sween, R., Deb, S., Carruth, D. W., Waddell, D., & Furuichi, M. (2016). Development of an effective pedestrian simulator for research. In *Advances in human aspects of transportation* (pp. 183–191). Springer International Publishing.
Urmson, C. P., Mahon, I. J., Dolgov, D. A., & Zhu, J. (2015). U.S. Patent No. 8,954,252. Washington, DC: U.S. Patent and Trademark Office.
Veruggio, G., Operto, F., & Bekey, G. (2016). Roboethics: Social and Ethical Implications. In *Springer handbook of robotics* (pp. 2135–2160). Springer International Publishing.
Zhang, J., Vinkhuyzen, E., & Cefkin, M. (2017). Evaluation of an autonomous vehicle external communication system concept: A survey study. In *International conference on applied human factors and ergonomics* (pp. 650–661). Cham: Springer.
Zito, G. A., Cazzoli, D., Scheffler, L., Jäger, M., Müri, R. M., Mosimann, U. P., ... Nef, T. (2015). Street crossing behavior in younger and older pedestrians: An eye- and head-tracking study. *BMC Geriatrics, 15*(1), 176.